2. If θ is an acute angle, then find

   (i) $\sin\left(\frac{\pi}{4}-\frac{\theta}{2}\right)$, when $\sin\theta=\frac{1}{25}$.

   (ii) $\cos\left(\frac{\pi}{4}+\frac{\theta}{2}\right)$, when $\sin\theta=\frac{8}{9}$.

3. If $\cos\theta=\frac{1}{2}\left(a+\frac{1}{a}\right)$, show that $\cos 3\theta=\frac{1}{2}\left(a^3+\frac{1}{a^3}\right)$.

4. Prove that $\cos 5\theta=16\cos^5\theta-20\cos^3\theta+5\cos\theta$.

5. Prove that $\sin 4\alpha=4\tan\alpha\frac{1-\tan^2\alpha}{\left(1+\tan^2\alpha\right)^2}$.

6. If $A+B=45^\circ$, show that $(1+\tan A)\,(1+\tan B)=2$.

7. Prove that $(1+\tan 1^\circ)(1+\tan 2^\circ)(1+\tan 3^\circ)\ldots(1+\tan 44^\circ)$ is a multiple of 4.

8. Prove that $\tan\left(\frac{\pi}{4}+\theta\right)-\tan\left(\frac{\pi}{4}-\theta\right)=2\tan 2\theta$.

9. Show that $\cot\left(7\frac{1}{2}^\circ\right)=\sqrt{2}+\sqrt{3}+\sqrt{4}+\sqrt{6}$.

10. Prove that $(1+\sec 2\theta)(1+\sec 4\theta)\ldots(1+\sec 2^n\theta)=\tan 2^n\theta\ \cot\theta$.

11. Prove that $32(\sqrt{3})\sin\frac{\pi}{48}\cos\frac{\pi}{48}\cos\frac{\pi}{24}\cos\frac{\pi}{12}\cos\frac{\pi}{6}=3$.

3.5.3 Product to Sum and Sum to Product Identities

Some applications of trigonometric functions demand that a product of trigonometric functions be written as sum or difference of trigonometric functions. The sum and difference identities for the cosine and sine functions look amazingly like each other except for the sign in the middle. So, we tend to combine them to get nice identities. Thus, we use them to derive several identities that make it possible to rewrite a product as a sum or a sum as a product.

We know that

$$\sin(A+B) = \sin A\cos B+\cos A\sin B \tag{3.1}$$
$$\sin(A-B) = \sin A\cos B-\cos A\sin B \tag{3.2}$$
$$\cos(A+B) = \cos A\cos B-\sin A\sin B \tag{3.3}$$
$$\cos(A-B) = \cos A\cos B+\sin A\sin B \tag{3.4}$$

From the above identities, we can easily derive the following Product to Sum identities.

$$\sin A\cos B = \frac{1}{2}[\sin(A+B)+\sin(A-B)] \tag{3.5}$$
$$\cos A\sin B = \frac{1}{2}[\sin(A+B)-\sin(A-B)] \tag{3.6}$$
$$\cos A\cos B = \frac{1}{2}[\cos(A+B)+\cos(A-B)] \tag{3.7}$$
$$\sin A\sin B = \frac{1}{2}[\cos(A-B)-\cos(A+B)] \tag{3.8}$$

The above identities are very important whenever need arises to transform the product of sine and cosine into sum. This idea is very much useful in evaluation of some integrals.

To get Sum to Product identities, let us introduce the substitutions $A+B=C$ and $A-B=D$ or equivalently $A=\frac{C+D}{2}, B=\frac{C-D}{2}$ in the product to sum identities (3.5) to (3.8). Then, we have the following Sum to Product identities

$$\sin C+\sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2} \tag{3.9}$$